🧪 Lab Assignment – Python Sets (Updated)

◆ Set Creation & Conversion

1. Create a set using `set()` from user-entered numbers (space-separated) and print only the unique values.
2. Create a set that contains a number, a decimal value, and a word. Print its length.
3. Convert a given list of city names with duplicates into a set and display the result.
4. Create an empty set and add three different elements entered by the user.

◆ Adding & Updating

5. Start with set `{2, 4, 6}`. Add one new number entered by the user and print the updated set.
6. Update a set using values from a tuple of integers.
7. Update a set using characters from a string entered by the user.
8. Merge two sets into one by updating the first set using the second.

◆ Removing & Membership

9. Attempt to delete a user-entered element from a set using `discard()` and print the set afterward.
10. Write a program that removes all elements smaller than 10 from a given set of numbers.
11. Check whether a user-entered word exists in a predefined set of programming languages.
12. Given a set of numbers, remove all values divisible by 3 and print the final set.

◆ Set Operations & Relations

13. For two given sets, display their:

- union
- intersection
- difference (A – B)
- symmetric difference

14. Given two sets representing students enrolled in Math and Physics, find students enrolled in both subjects.

15. From two sets of roll numbers, find students enrolled in **exactly one course**.

16. Write a program that checks whether two given sets have **no common elements**.

17. Given two sets A and B, print whether A is a **subset of B** and whether A is a **superset of B**.

Optional Challenge

18. Take a sentence from the user and print the set of unique characters used in it (ignore spaces).